

Similarity & Proportional Reasoning

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **determine** image lengths and coordinates under a dilation.

RELATED SKILLS

☐ D Calculate and interpret a similarity or dilation scale factor.☐ A Match corresponding vertices, sides, and angles in order.

☐ 2. I can **identify** the scale factor of a dilation or similarity relationship.

RELATED SKILLS

☐ I Recall and correctly use the terms similarity, dilation, scale factor, and corresponding parts.☐ D Calculate and interpret a similarity or dilation scale factor.☐ A Use cross multiplication when two ratios form a proportion.

☐ 3. I can **write** a similarity statement with corresponding vertices in the correct order.

RELATED SKILLS

☐ I Write similarity statements in corresponding order.☐ D Match corresponding vertices, sides, and angles in order.

☐ 4. I can **use** proportional corresponding sides to solve similar-polygon problems.

RELATED SKILLS

☐ I Set up proportions using corresponding sides.☐ A Match corresponding vertices, sides, and angles in order.☐ A Use cross multiplication when two ratios form a proportion.

☐ 5. I can **relate** similarity scale factor to perimeter and area scale factors.

RELATED SKILLS

☐ I Relate length scale factor to perimeter and area factors.☐ A Calculate and interpret a similarity or dilation scale factor.

☐ 6. I can **prove** triangles similar by AA, SSS, or SAS.

RELATED SKILLS

☐ I Recognize AA triangle similarity.☐ I Test SSS or SAS triangle similarity.☐ D Organize statements and reasons into a valid proof sequence.☐ A Write similarity statements in corresponding order.

☐ **7. I can **apply** triangle proportionality theorems and their converses.**

RELATED SKILLS

I Apply triangle proportionality relationships.

A Recognize and apply the converse of a conditional statement.

☐ **8. I can **solve and justify** an indirect-measurement or other similarity application.**

RELATED SKILLS

D Set up proportions using corresponding sides.

A Calculate and interpret a similarity or dilation scale factor.

A Select a theorem that connects known facts to a target statement.
